



mec

SOFTWARE DEVELOPMENT AND SIGNAL PROCESSING FOR HIGH THROUGHPUT SILICON MULTIELECTRODE ARRAY SYSTEMS

LIFE SCIENCE TECHNOLOGY DEPARTMENT

TEAM CELL AND TISSUE TECHNOLOGY

Shuo-Wen, Chang

Supervisor: Dr. Beatrice Miccoli

TEAM MEMBER INTRODUCTION



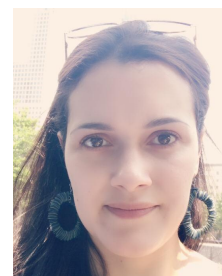
Beatrice
Miccoli



Laurien Mols



Pieris Zindilis



Dr. C.M. Lopez



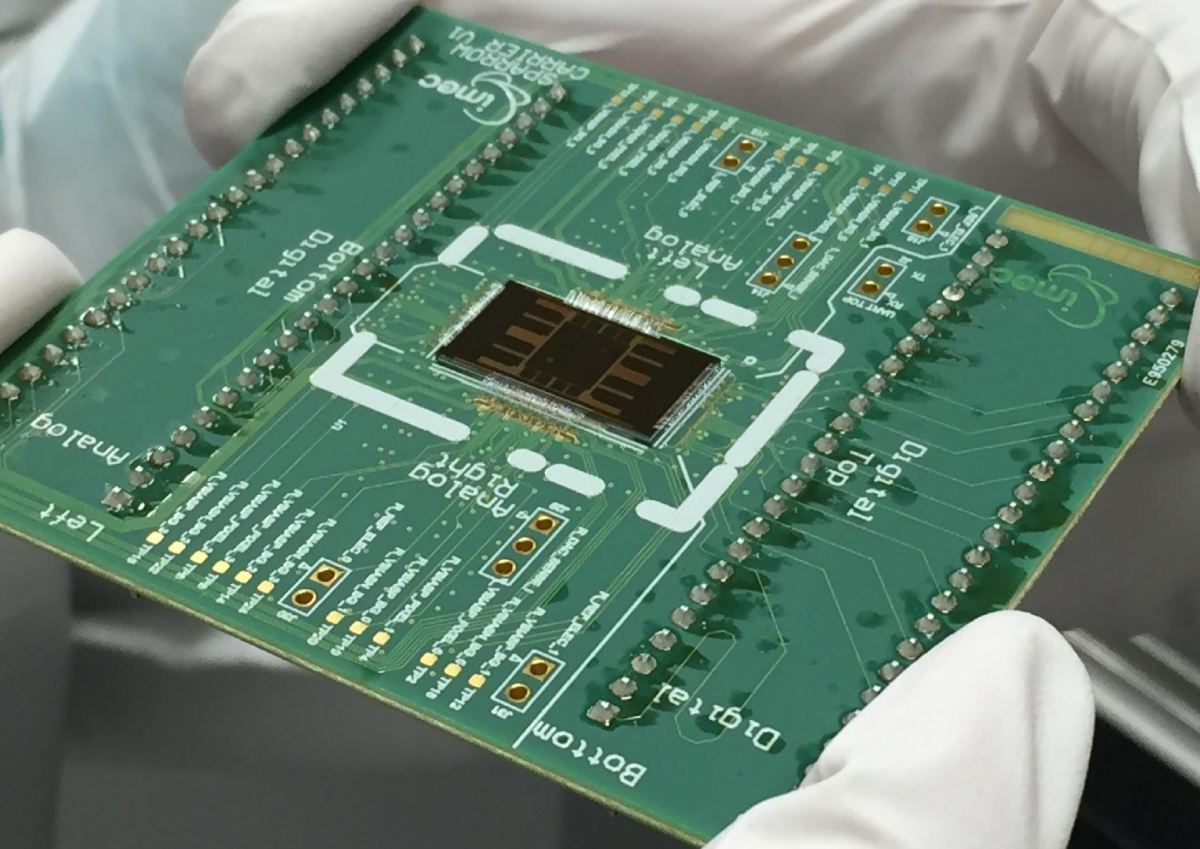
CLINICAL DEMAND

Two main goals of our team:

Goal 1 : Fundamental biology research

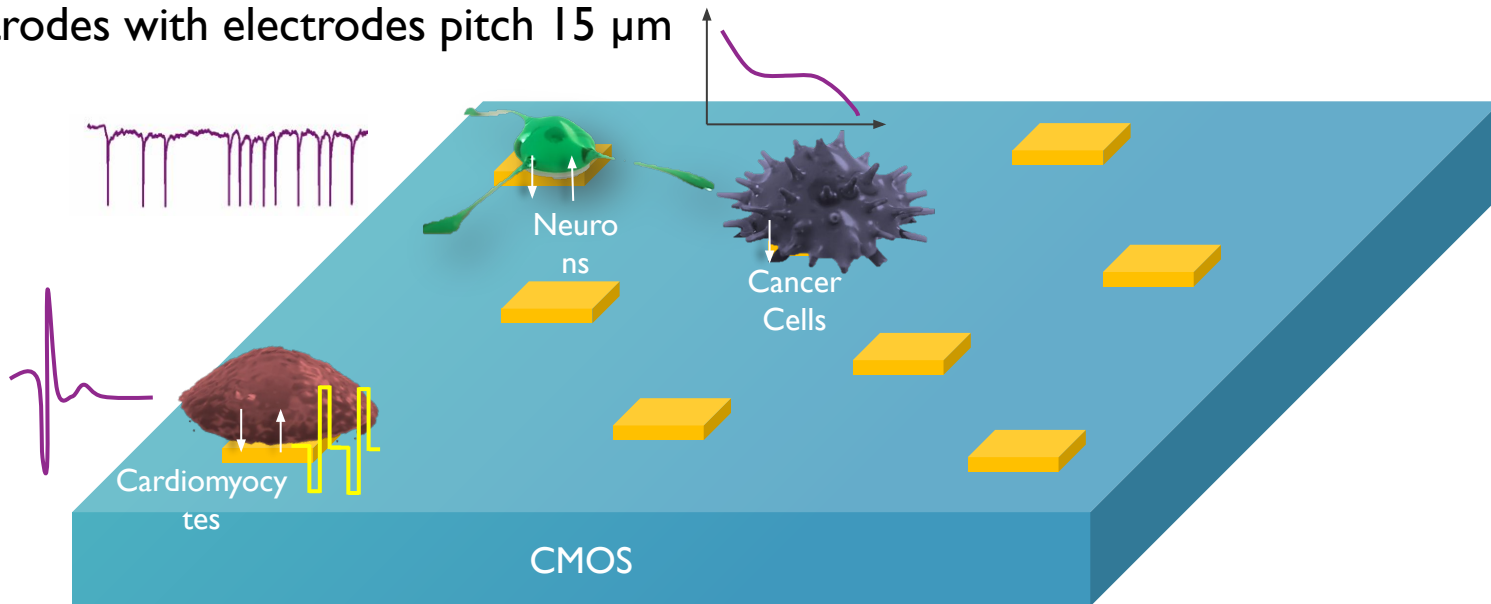
Goal 2 : Drug development and
pharmaceutical industry

SPARROW MEA SYSTEM

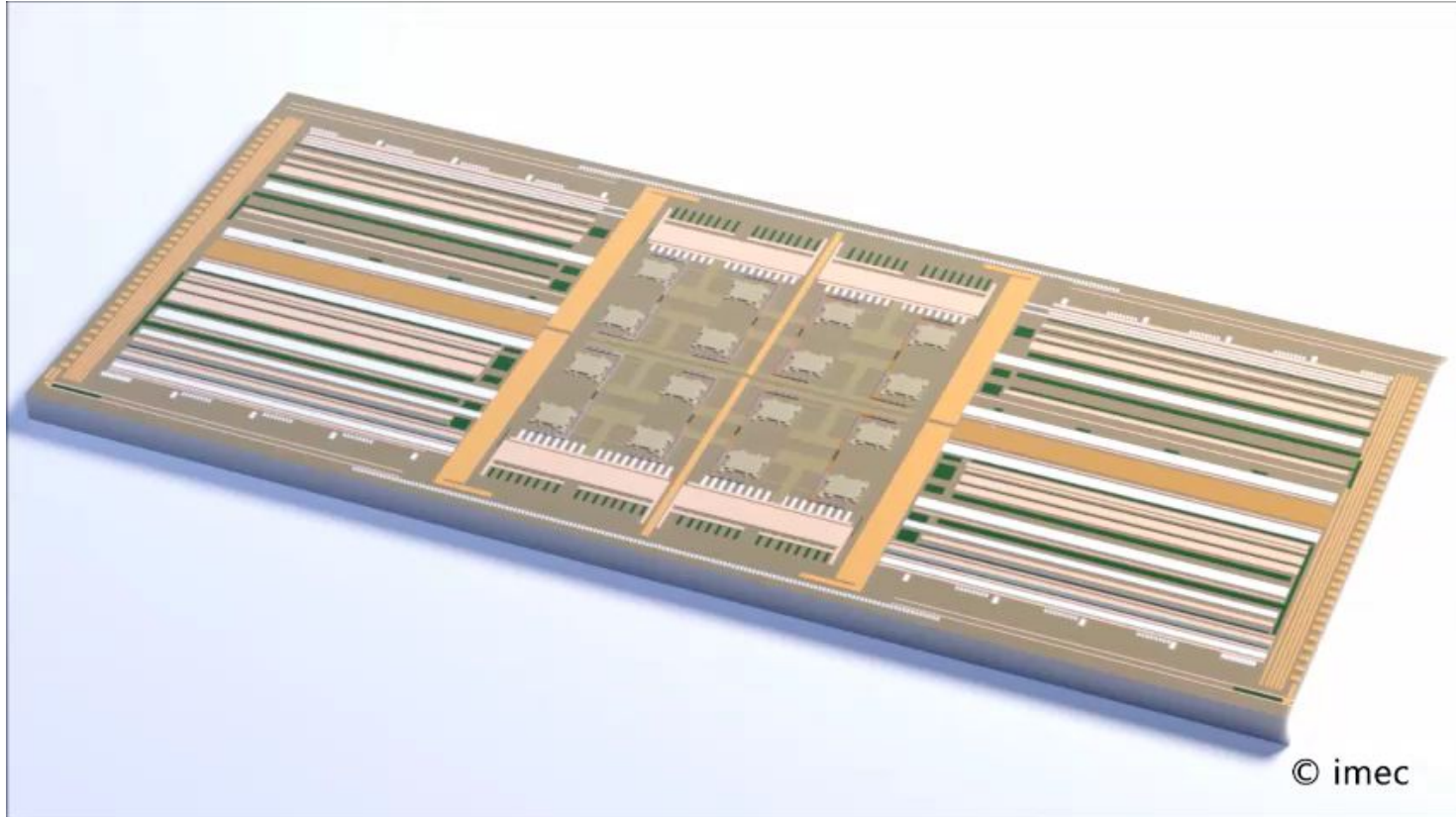


MAIN ACTIVITY

- Imec designs and makes breakthrough CMOS-based Multi Electrode Arrays (MEAs).
- By introducing CMOS chip technology in MEAs, imec enables high-quality, high-throughput data acquisition for cell interaction and behavior.
- 16k electrodes with electrodes pitch 15 μm



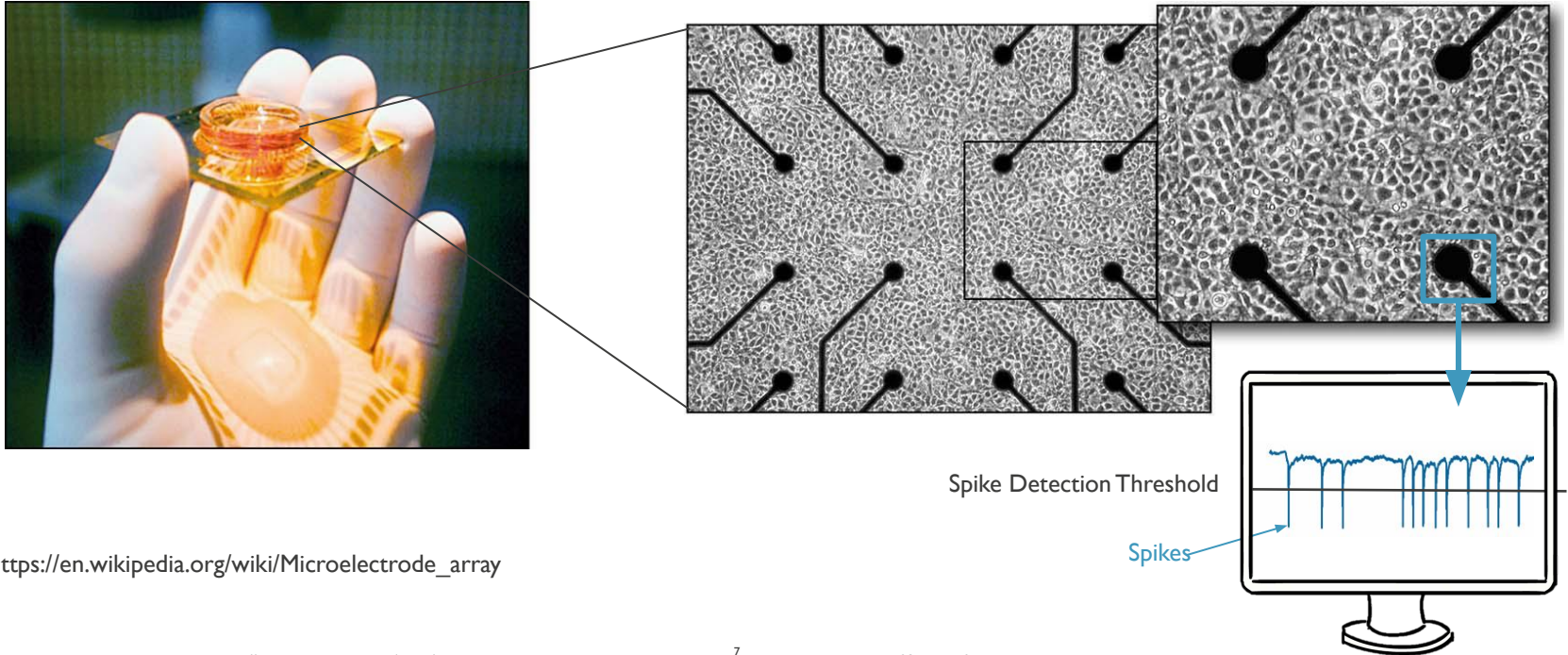
SYSTEM OVERVIEW



MAIN GOAL

- Develop chips for the study of cells *in vitro*

This is the reason imec fabricated a multi electrode array (MEA) chip



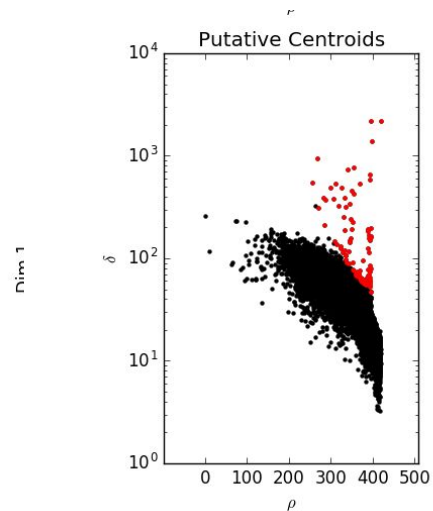
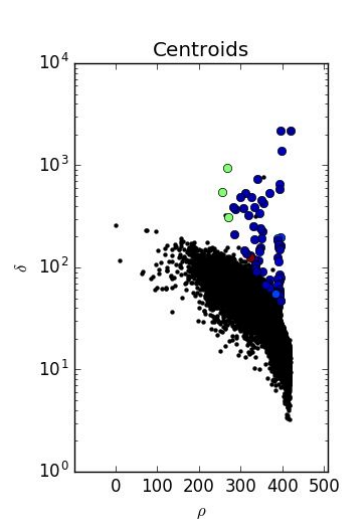
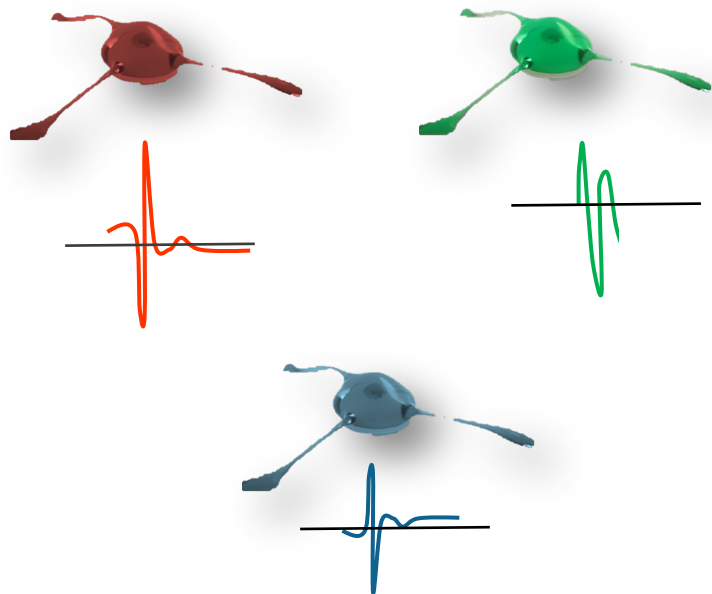
https://en.wikipedia.org/wiki/Microelectrode_array

MY PROJECT A GUI (GRAPH USER INTERFACE)



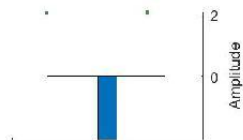
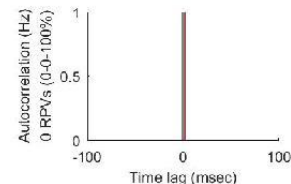
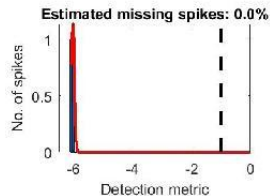
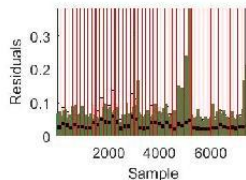
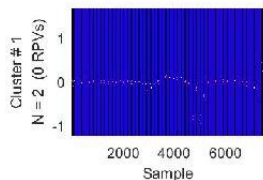
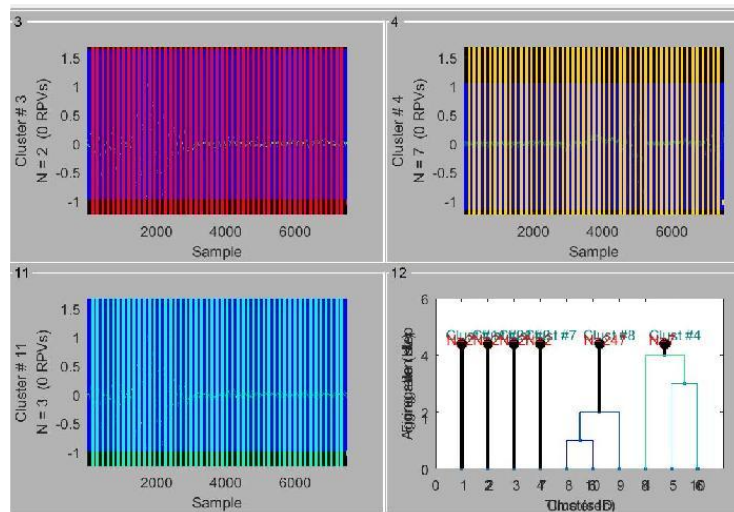
Multi signal plot generated by Spyking-Cirrus

CELL CLUSTERING RESULTS



Cluster plot generated by UltraMegaSort

PRELIMINARY RESULTS



Cluster plot generated by UltraMegaSort

TEAM'S PUBLICATION RESULTS

Frontiers in Neuroscience (h5-index: 70; h5-median: 102) Journal Paper

- **Accepted** in Dec, 2018.
- Beatrice M., Carolina M. Lopez, Erkuden Goikoetxea, Jan Putzeys, Makrina Sekeri, Olga Krylychkina, **Shuo-Wen Chang**, Andrea Firrincieli, Alexandru Andrei, Veerle Reumers, Dries Braeken, "[High-Density Electrical Recording and Impedance Imaging with a Multi-Modal CMOS Multi-Electrode Array Chip](#) "
- Main contribution: Perform CNN to classify the image of microscopy cells image.

The screenshot shows the Frontiers Review Forum interface. At the top, there's a navigation bar with links: HOME, ABOUT, SUBMIT, JOURNALS, RESEARCH TOPICS, and a search icon. On the right, it says 'MY FRONTIERS' with a circular icon, a bar chart icon, and the user name 'SHUO-WEN CHANG' next to a profile icon. Below this is a red banner for 'REVIEW FORUM' with 'Co-author' and a link '? Need Help? Contact us'. A progress bar shows seven steps: 1. Initial Validation (checked), 2. Editorial Assignment (checked), 3. Independent Review (checked), 4. Interactive Review (active), 5. Review Finalized, 6. Final Validation, and 7. Final Decision. The main content area displays the paper title 'High-Density Electrical Recording and Impedance Imaging with a Multi-Modal CMOS Multi-Electrode Array Chip', with 'CMOS Multi-Electrode Array Chip' highlighted in a blue box. Below the title, the authors are listed: Beatrice Miccoli, Carolina Mora Lopez, Erkuden Goikoetxea, Jan Putzeys, Makrina Sekeri, Olga Krylychkina, Shuo-Wen Chang, Andrea Firrincieli, Alexandru Andrei, Veerle Reumers and Dries Braeken. It also states 'Original Research, Front. Neurosci. - Neural Technology', 'Submitted on: 14 Dec 2018, Edited by: Udo Kraushaar', and 'Manuscript ID: 443106'. The research topic is 'Microelectrode Arrays - Monitoring and Manipulating Cells and Networks over Extended Scales at Highest Resolutions', and keywords include 'Multi-Modal, Impedance imaging, hippocampal neurons, CMOS, high density (HD), Multielectrode array (MEA)'. On the right side of the paper details, there are three icons with links: 'Download latest manuscript', 'View submitted files history', and 'View Invoice'.



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